

10(a)	any four from: 1 troponin binds Ca^{2+} and changes shape ; 2 (so) tropomyosin moves away from the binding sites on actin ; 3 myosin heads, bind to / form cross-bridges, with actin ; 4 myosin head, pulls actin / performs power stroke, to, either shorten / contract, sarcomere or cause Z lines to move closer together / Z lines move closer to M lines ; 5 myosin head, is ATPase / binds to ATP, to allow detachment ;	4
10(b)	any three from: 1 succinylcholine binds to ACh receptors or succinylcholine acts as a competitive inhibitor ; R if binds to active site of receptor 2 (sodium ions) channel proteins do not open ; R if voltage gated 3 sodium ions do not enter, muscle cell / sarcoplasm ; 4 sarcolemma / post-synaptic membrane, not depolarised ;	3

Question	Answer	Marks
10(c)	any three from: 1 sex-linked / gene on X chromosome ; 2 recessive (allele) ; 3 males only have one, copy of gene / allele / X chromosome ; 4 female heterozygotes, do not have DMD / are carriers or only homozygous females have DMD ;	3